

Trigonometric $(\sin(\heartsuit))' \rightarrow \cos(\heartsuit) \cdot \heartsuit'$

$$(\cos(\heartsuit))' \rightarrow -\sin(\heartsuit) \cdot \heartsuit'$$

$$\sin(\sqrt{x}) \rightarrow \cos(\sqrt{x}) \cdot (\sqrt{x})' = \cos(\sqrt{x}) \cdot \frac{1}{2\sqrt{x}}$$

Exponential $(e^{\heartsuit})' \rightarrow e^{\heartsuit} \cdot \heartsuit'$

$$e^{5-3x^2} \rightarrow e^{5-3x^2} \cdot (5-3x^2)' = -6x \cdot e^{5-3x^2}$$

Logarithmic $(\ln \heartsuit)' \rightarrow \frac{1}{\heartsuit} \cdot \heartsuit'$

$$\ln(\cos(x)) \rightarrow \frac{1}{\cos(x)} \cdot (\cos(x))' = \frac{1}{\cos(x)} \cdot (-\sin(x)) = -\tan(x)$$

Powers : $(\heartsuit^h)' \rightarrow h \cdot \heartsuit^{h-1} \cdot \heartsuit'$

$$\frac{2}{\sqrt[3]{5-x^2}} = 2 \cdot (5-x^2)^{-1/3} \rightarrow 2 \cdot (-1/3) \cdot (5-x^2)^{-4/3} \cdot (5-x^2)'$$

$$\frac{4}{3} x \cdot (5-x^2)^{-4/3} = \frac{4x}{3 \cdot \sqrt[3]{(5-x^2)^4}}$$

$$f(x) = x \cdot \underbrace{|1-x|}_0 \quad f(x) = \begin{cases} x \cdot (-(1-x)) & ; x \geq 1 \\ x \cdot (1-x) & ; x < 1 \end{cases}$$

$$f(x) = \begin{cases} x^2 - x & ; x \geq 1 \\ -x^2 + x & ; x < 1 \end{cases} \quad \left. \begin{array}{l} \lim_{x \rightarrow 1^+} f(x) = 1^2 - 1 = 0 \\ \lim_{x \rightarrow 1^-} f(x) = -1^2 + 1 = 0 \end{array} \right\} = f(1)$$

$$f'(x) = \begin{cases} 2x - 1 & ; x \geq 1 \\ -2x + 1 & , x < 1 \end{cases} \quad \left. \begin{array}{l} \lim_{x \rightarrow 1^+} f'(x) = 1 \\ \lim_{x \rightarrow 1^-} f'(x) = -1 \end{array} \right\} \neq$$

nicht differenzierbar